1 Construct comparable perturbation responses

EXPERIMENTAL CONTEXTS

Blood panel

10 species; LPS dose and time

TG-GATEs

Human / rat; culture / liver

Perturbed − Control

Align genes within each dataset

Keep subjects and conditions identified

Response matrix

Rows: subjects within conditions

Genes

Response (a.u.)

−1.5 0.0 1.5

Blood panel

TG-GATEs

2 Quantify between-species redundancy

Average subjects within species and condition

Species means

A
B
C
D

Aligned genes

Pearson

across genes

Redundancy ρ

Species pairs

Split subjects; rebuild both matrices

Compare the same species-pair ranks

Reliability is established before interpreting transfer.

3 Calibrate agreement using replicates

A / B = a time, species or system boundary

A1

B1

rel A

All cross-side pairings

rel B

A2

B2

$$\text{Fraction of ceiling} = \frac{\text{observed}}{\sqrt{\text{rel}_A \, \text{rel}_B}}$$

Independent controls; compare within ceiling bands

Check reliability | Inspect shared controls when ratios exceed 1 | Re-test context transfer

4 Interpret what transfers, and where it stops

Panel design

Donor value · Fig. 2

Context reuse

Time / tissue · Fig. 3

Response specificity

Shared / residual · Fig. 4

Boundary comparison

Ceiling

Fraction of ceiling · Table 2

All matrices and output miniatures are illustrative; empirical results appear in the indicated figures and table.